

| 🔊 WAVSYNTH PARAMETERS |                                                                                 |
|-----------------------|---------------------------------------------------------------------------------|
| SHAPE 00—03           | Pulse 12%, 25%, 50%, 75%<br><i>Two-level waves: clean even at 8 bits</i>        |
| SHAPE 04—08           | Saw, triangle, sine, pitched noise, noise<br><i>09 and up are wavetables</i>    |
| SIZE                  | Waveform length<br><i>Also sets how smooth PWM sounds</i>                       |
| MULT                  | Repeats the shape<br><i>Sounds like hard sync as it rises</i>                   |
| WARP                  | Skews the shape<br><i>Pushes it to one side of the table</i>                    |
| SCAN                  | Mirror or scan<br><i>Shapes 00—08: mirror point (PWM); wavetables: 64 waves</i> |
| WAV LP/HP             | 8-bit filters<br><i>WAV LP, HP, BP, BS reshape the wave itself</i>              |
| FX                    | OSC · SIZ · MUL · WRP · SCN<br><i>Automate each control per step</i>            |

| ☰ WAVETABLES 09—45 |                                                                      |
|--------------------|----------------------------------------------------------------------|
| 09—1A              | OSC<br><i>CRUSH, FOLDING, FUZZY, GHOST, LIQUID, TUBE, WAVES...</i>   |
| 1B—21              | BNK<br><i>DRIP, FROGGY, INSONIC, RADIUS, SCRATCH, SMOOTH, WOBBLE</i> |
| 22—2E              | HRM<br><i>GENTLE, HARMONIC, PLAITS01—02, TONAL, TWINE...</i>         |
| 2F—43              | EFX<br><i>ALIEN, FORMANT, RINGS, SHIMMER, SPOOKY, VOCAL, ZAP...</i>  |
| 44—45              | VOX<br><i>BRAIDS, VOXSYNTH</i>                                       |

| ⚙️ GOOD TO KNOW                                                                                                                                                                                                        |             |
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| 8-Bit Character                                                                                                                                                                                                        | 256 LEVELS  |
| <ul style="list-style-type: none"><li>Wavsynth uses 256 levels, so saws look like staircases</li><li>Those edges add bright, small-speaker harmonics</li><li>Pulse waves have only two levels and stay clean</li></ul> |             |
| Same Tables in FM                                                                                                                                                                                                      | FM W SHAPES |
| <ul style="list-style-type: none"><li>FM Synth operators can use these wavetables too</li><li>There, FB scans the table instead of adding feedback</li></ul>                                                           |             |

| 📺 WAVSYNTH RECIPES                                                                                                                                                                                                       |                  |
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| Chip Pulse Bass                                                                                                                                                                                                          | PULSE 50%        |
| <ul style="list-style-type: none"><li>SHAPE PULSE 50%: only two levels, so 8-bit costs nothing</li><li>Short AHD on volume; WARP to change its hollow tone</li><li>A little MFX send (chorus) widens it</li></ul>        |                  |
| PWM Lead                                                                                                                                                                                                                 | SCAN             |
| <ul style="list-style-type: none"><li>SHAPE PULSE 50%, then an LFO on SCAN sweeps the width</li><li>Larger SIZE makes the sweep smoother</li></ul>                                                                       |                  |
| Chip Chords                                                                                                                                                                                                              | ARP 47           |
| <ul style="list-style-type: none"><li>PULSE 25% with ARP 47 (major) or ARP 37 (minor)</li><li>ARC sets the arpeggio mode and speed in ticks</li><li>PVB after the attack for a classic wobbly lead</li></ul>             |                  |
| Noise Drums                                                                                                                                                                                                              | NOISE            |
| <ul style="list-style-type: none"><li>SHAPE NOISE with a very short AHD for hats and snares</li><li>PITCHED NOISE plays in tune: the note sets its color</li><li>Fast PBN down on pitched noise for laser zaps</li></ul> |                  |
| Wavetable Morph                                                                                                                                                                                                          | SCN              |
| <ul style="list-style-type: none"><li>Pick a wavetable (e.g. 26 HRM:HARMONIC or 3F EFX:VOCAL)</li><li>An LFO or table steps on SCAN morph through its 64 waves</li></ul>                                                 |                  |
| Sync-Style Sweeps                                                                                                                                                                                                        | MULT             |
| <ul style="list-style-type: none"><li>AHD on MULT stacks repeats of the shape as it rises</li><li>Sounds like a hard-sync sweep on leads and basses</li></ul>                                                            |                  |
| Tame or Crunch                                                                                                                                                                                                           | WAV LP · CUTOFF  |
| <ul style="list-style-type: none"><li>8-bit saws step like a staircase and add bright edges</li><li>A normal low-pass tames them; WAV filters add crunch</li></ul>                                                       |                  |
| Duty-Cycle Switching                                                                                                                                                                                                     | OSC 00—03        |
| <ul style="list-style-type: none"><li>OSC on a step switches between the four pulse widths</li><li>12% is thin and nasal, 50% is hollow and round</li></ul>                                                              |                  |
| Chip Kick                                                                                                                                                                                                                | SINE · AHD PITCH |
| <ul style="list-style-type: none"><li>SHAPE SINE with a short AHD on volume</li><li>A fast AHD on pitch drops it for the thump</li></ul>                                                                                 |                  |
| Echo Channel                                                                                                                                                                                                             | DEL · V          |
| <ul style="list-style-type: none"><li>Copy the lead phrase to a free track with lower V values</li><li>DEL on its first row delays it for a classic tracker echo</li></ul>                                               |                  |

| 🔥 HYPERSYNTH PARAMETERS |                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------|
| CHORD                   | 16 banks, up to 6 intervals<br><i>Each note is two saws; SHIFT + OPT copies, SHIFT + EDIT pastes</i> |
| SCALE                   | Song scale for the chord<br><i>00 follows the song or track scale</i>                                |
| SHIFT                   | Crossfade<br><i>Between intervals 1—3 and 4—6</i>                                                    |
| SWARM                   | Detune<br><i>Spreads each note's pair of saws</i>                                                    |
| WIDTH                   | Stereo spread<br><i>00 mono to FF full stereo</i>                                                    |
| SUBOSC                  | Square sub<br><i>Below 80: two octaves down · above 80: one</i>                                      |
| FX                      | CRD · SWM · WID · SUB<br><i>Change chords and tone per step</i>                                      |

| 🎵 HYPERSYNTH RECIPES                                                                                                                                                                                                  |               |
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| Lush Pad                                                                                                                                                                                                              | SWARM · WIDTH |
| <ul style="list-style-type: none"><li>CHORD: a triad or 7th (e.g. 0, 4, 7, 11)</li><li>Moderate SWARM, WIDTH toward FF, MFX and REV sends</li><li>ADSR on volume, slow ATK and long REL; end notes with OFF</li></ul> |               |
| Chord Progressions                                                                                                                                                                                                    | CRD           |
| <ul style="list-style-type: none"><li>Build a bank for each chord shape you need</li><li>CRD on a step switches banks; the note sets the root</li></ul>                                                               |               |
| Evolving Extensions                                                                                                                                                                                                   | SHIFT         |
| <ul style="list-style-type: none"><li>Intervals 1—3: a triad · 4—6: its 7th, 9th and 11th</li><li>A slow LFO on SHIFT morphs from plain to rich</li></ul>                                                             |               |
| Supersaw Stab                                                                                                                                                                                                         | SWARM · AHD   |
| <ul style="list-style-type: none"><li>High SWARM, WIDTH FF, short AHD on volume</li><li>A DEL send for trance-style echoes</li></ul>                                                                                  |               |
| Bass + Chord in One                                                                                                                                                                                                   | SUBOSC        |
| <ul style="list-style-type: none"><li>Raise SUBOSC for the root under the chord</li><li>Low CUTOFF and a short AHD on it for a stab</li></ul>                                                                         |               |
| Stay in Key                                                                                                                                                                                                           | SCALE         |
| <ul style="list-style-type: none"><li>Set SCALE to a song scale so intervals follow the key</li><li>SCA or SCG changes the scale mid-song</li></ul>                                                                   |               |
| Solid Low End                                                                                                                                                                                                         | WIDTH         |
| <ul style="list-style-type: none"><li>Keep WIDTH low when SUBOSC or low notes carry the bass</li><li>Save full width for pads that sit above it</li></ul>                                                             |               |
| Pluck Chords                                                                                                                                                                                                          | CUTOFF · AHD  |
| <ul style="list-style-type: none"><li>Low SWARM, short AHD on CUTOFF with some RES</li><li>Play off-beat chords for house and garage stabs</li></ul>                                                                  |               |